

Q1: Explain the meaning printed out in serial monitor after IMU sensing in Part1. Show how to change sampling rate to 50Hz and get 100 data points.

- **ts (Timestep)**: 資料的採樣序號，從 0 開始遞增。
- **t_us (Time in us)**: 取得樣本時的系統微秒時間。
- **ax, ay, az**: MPU6050 沿 X、Y、Z 軸量測到的**原始加速度計**讀數（未經轉換的 16-bit 有號數值）。
- **gx, gy, gz**: MPU6050 沿 X、Y、Z 軸量測到的**原始陀螺儀**讀數（角速度，16-bit 有號數值）。

```
#define SAMPLE_RATE_HZ 50
#define SAMPLE_INTERVAL_US 20000UL
#define RECORD_SECONDS 2.0
#define NUM_SAMPLES 100
```

$$f = 50\text{HZ}$$

$$T = 20000\mu s$$

$$\text{總時間為 } 20ms \times 100\text{個} = 2s$$

$$\text{取樣數} = 100$$

Q2: For preprocess_input(): take a screenshot of the serial monitor and circle one raw value you want to observe. Trace through the three steps for this sample value. Show the intermediate float values after each step and the final INT8 result, and verify the output is within [-128, 127].

```
t= 50  RAW |  -2980  -1348  18712  -512  -32768  -5100
        PRE |      -6    -13     12    -1    -128    -8
```

```
=== Step-by-Step Trace for gy = -32768 ===
Step 1 (Physical)      : -32768 / 131.0 = -250.13740
Step 2 (Standardize)  : (-250.13740 - 8.56773) / 38.42800 = -6.73220
Step 3 (Quantize)     : -6.73220 / 0.0501416 = -134.26378
                     (Round) : round(-134.26378) = -134
Final INT8 (Clamp)    : min(max(-134, -128), 127) = -128
```

- 針對陀螺儀 gy 解析，最後有clamp。

Q3:For requantize_int8() (Todo 3.2): explain why int64_t is required for the intermediate product instead of int32_t. Provide a concrete numerical example using actual mult and shift values defined in nn_weights.h with an acc picked by you, showing that the product would overflow a 32-bit type.

(1):

- 32-bit 的累加值 (acc) 乘上一個 32-bit 的乘數 (mult)，其結果最多可能會需要 64 個位元 (64-bit) 才能完整儲存。

(2):

```
43  #define RQ_MULT_DENSE1 1940942
44  #define RQ_SHIFT_DENSE1 30
```

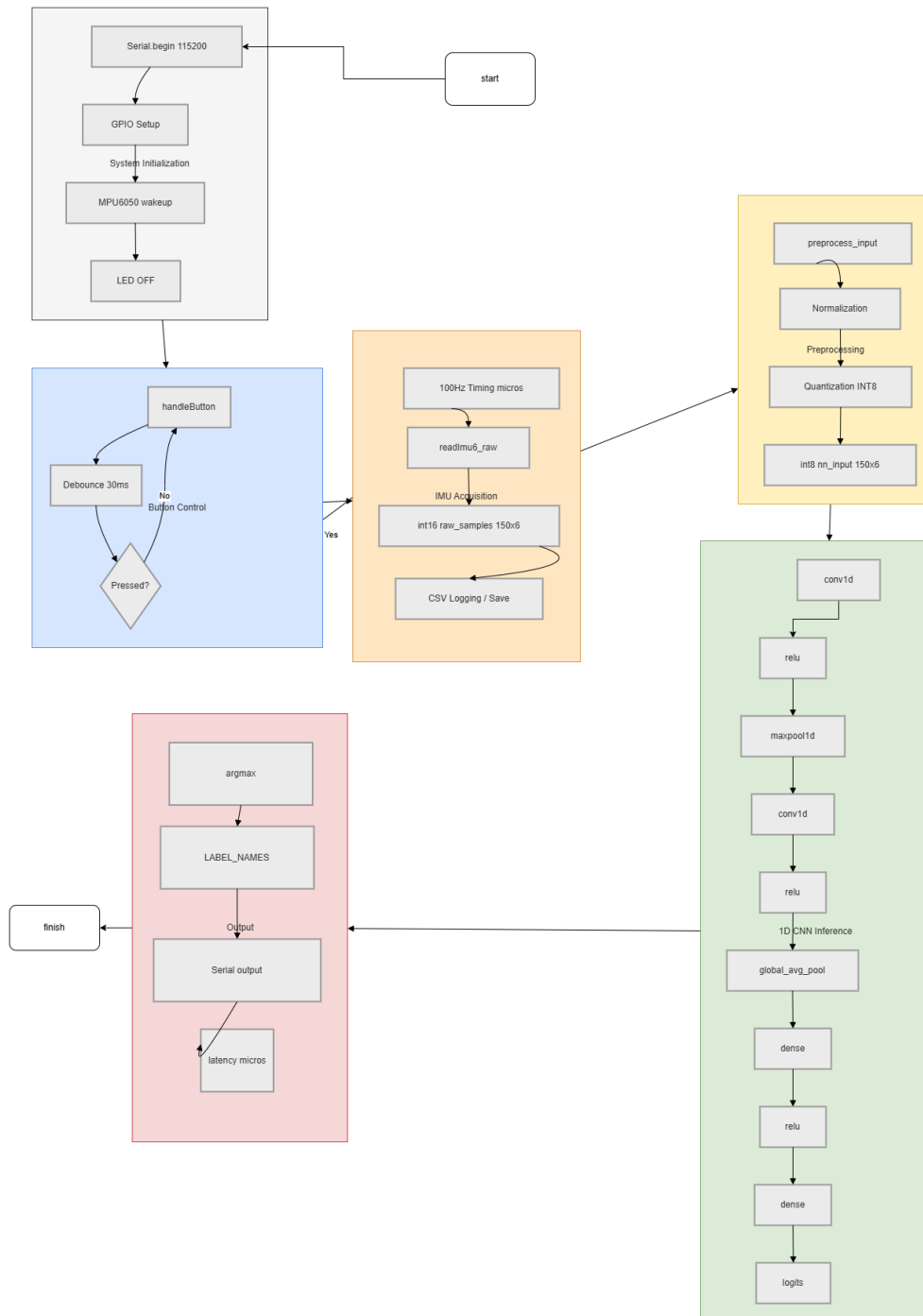
- 若挑選acc為2000， $2000 \times 1940942 = 3881884000$ 這數已經遠大於 2,147,483,647。不轉型會變成-413,083,296
- 因此要先轉64bit才能做位移。

Q4 Measure and report the latency of the whole process, starting from readlm6_raw() to the last dense() function, by using micros().

```
# PROCESS_LATENCY_US,1561031
# INFERENCE_LATENCY_US,57267
# PREDICTION,updown
# END
```

約等於採樣時間加上NN推論時間。

Q5: Draw the flowchart of the whole system, starting from collecting IMU data, preprocessing to inference. Details please refer to [System Flowchart 25%] section in page 7



Q6:Other interesting works or observations.

在做手勢辨識時不小心將MPU6050斷電1ms，會導致數據都是錯的，要將他重新喚醒。